

Jun-Kun Chen

Ph.D. Candidate in Computer Science | Expected Dec. 2026

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Legal name: Junkun Chen | Publications: Jun-Kun Chen / J.-K. Chen

Research Focus

Research in **diffusion-based controllable image/video generation**, with first/co-first-author work at NeurIPS and CVPR and prior research internships at Meta and SpreeAI. Contributions span **multimodal conditioning**, image/video training curricula, and long-video consistency, with current work on visual world models and multimodal post-training.

Industry Experience

Meta Reality Labs

Research Scientist Intern (2025 continued part-time Aug.–Dec.)

May–Dec. 2025; May–Aug. 2023
Bay Area, CA, U.S. / Zurich, Switzerland

- **2025:** Designed parallel training workflows for 3D/video generation at **128–256-GPU scale** across multi-node clusters.
- **2023:** Developed multi-view conditioning, structured noise, and self-supervised consistency training for diffusion editing; published as *ConsistDreamer* at CVPR 2024.

SpreeAI

AI Research Intern

May 2024 – May 2025
Remote, U.S.

- **Dress&Dance:** designed **multimodal conditioning** and progressive image/video training for controllable 5-second try-on videos at **1152×720, 24 FPS**.
- **Virtual Fitting Room** (NeurIPS 2025): extended this work to **minute-scale, identity-consistent generation** with anchored autoregression; evaluated up to 90 seconds using short-clip training.

Selected Publications

1. Virtual Fitting Room: Generating Arbitrarily Long Videos of Virtual Try-On from a Single Image  **NeurIPS 2025**
Jun-Kun Chen, Aayush Bansal, Minh Phuoc Vo, Yu-Xiong Wang [Project](#) | [Early preview \(arXiv, Sep. 2025\)](#)
2. ProEdit: Simple Progression is All You Need for High-Quality 3D Scene Editing  **NeurIPS 2024**
Jun-Kun Chen, Yu-Xiong Wang [Project](#) | [arXiv \(Nov. 2024\)](#)
3. ConsistDreamer: 3D-Consistent 2D Diffusion for High-Fidelity Scene Editing  **CVPR 2024**
Jun-Kun Chen, Samuel Rota Bulò, Norman Müller, Lorenzo Porzi, Peter Kotschieder, Yu-Xiong Wang [Project](#) | [arXiv \(Jun. 2024\)](#)
4. Instruct 4D-to-4D: Editing 4D Scenes as Pseudo-3D Scenes Using 2D Diffusion  **CVPR 2024**
Linzhao Mou*, **Jun-Kun Chen***, Yu-Xiong Wang [Project](#) | [arXiv \(Jun. 2024\)](#)
5. HDEdit: Editing Videos and 3D Scenes with Video Diffusion Through Hierarchical Task Decomposition  **3DV 2026**
Yanming Zhang*, **Jun-Kun Chen***, Jipeng Lyu, Yu-Xiong Wang [Project](#) | [Earlier version \(arXiv, Mar. 2025\)](#)
6. Dress&Dance: Dress up and Dance as You Like It  **WACV 2027**, under review
Jun-Kun Chen, Aayush Bansal, Minh Phuoc Vo, Yu-Xiong Wang [Project](#) | [Early preview \(arXiv, Aug. 2025\)](#)

*Equal contribution. Earlier publications also appear as Junkun Chen / J. Chen.

Current Research: Generative World Models

Research direction: connecting language reasoning and visual generation through training grounded in predicted outcomes.

Computer Use — Action-Conditioned Image-Diffusion World Model

- **Post-trained image diffusion** on SenseNova MoT for next-screen prediction conditioned on the current screenshot and action.
- Designed **action-consequence and content-preservation supervision**; incorporated interaction traces and failure cases into Reasoner training.

Physics — Video-Diffusion World Model for Physical Events

- Adapted **Cosmos3 video diffusion** for physical-event prediction from observed RGB history.
- Implemented **online teacher distillation**, training the language Reasoner and diffusion Generator from video prediction errors.

Robotics — Motion-Conditioned Video Diffusion World Model

- Built an initial synthetic **video-diffusion prototype** for robot–object interaction prediction from a scene and prescribed motion.
- Designed **visual-proxy and paired-correspondence conditioning** to represent motion across robot–world setups.

Education

University of Illinois Urbana-Champaign

Ph.D. Candidate in Computer Science; Advisor: Prof. **Yu-Xiong Wang**

2021 – Dec. 2026 (expected)
Urbana, IL

Tsinghua University

B.Eng. in Computer Science, Yao Class (IIIS); Major GPA: 3.94/4.0

2017 – 2021
Beijing, China

Technical Skills & Honors

Tools & Methods: PyTorch; Python; C++; image/video diffusion; diffusion Transformers (DiT); LoRA; vision-language models; multimodal conditioning; training-data construction; benchmark design; quantitative and human evaluation; diffusion post-training.

Competition Highlights: ICPC EC-Final Gold (4th); ICPC Qingdao Gold (2nd); CCPC Harbin Gold; NOI Gold.

Selected Research Contributions

Controllable Try-On to Long-Horizon Video — Dress&Dance and VFR

- **Dress&Dance**: designed attention-based **multimodal conditioning** and a progressive image/video training curriculum; generated 5-second try-on videos at 1152×720 and 24 FPS with garment, identity, and motion control.
- Constructed garment/user/video training triplets using **synthetic appearance conditions with real-video targets**; staged the conditioning and resolution curricula to teach new controls while retaining pretrained visual priors.
- **Virtual Fitting Room**: designed **anchor/prefix conditioning** and immediate refinement for long-video generation from short training clips; scored higher on VBench subject/background consistency and motion smoothness than D&D+FramePack and D&D+Kling 2.0 pipelines on 30–50 s hard-motion clips.

General-Purpose Diffusion Editing — HDEdit and ProEdit

- **HDEdit**: designed **hierarchical task decomposition** that separates instruction fulfillment from content preservation, combining LLM-based scheduling with interpolation-based refinement for training-free video and scene editing.
- **ProEdit**: developed tuning-free preservation control and a **progressive editing workflow** with preview and edit-strength control, enabling high-fidelity geometry and appearance editing.

Multi-View and Temporal Consistency — ConsistDreamer and Instruct 4D-to-4D

- **ConsistDreamer**: created context-rich multi-view conditioning, **3D-consistent structured noise**, and self-supervised warping consistency for high-fidelity diffusion editing across views.
- **Instruct 4D-to-4D**: extended 2D diffusion to long-term dynamic scenes through **anchor-aware attention** and optical-flow-guided sliding windows.

Geometric Visual Representations — NeuralEditor and PointTree

- Built point-based radiance-field and relaxed k-d-tree representations for editable scene geometry and **transformation-robust point-cloud encoding**.

Additional Publications

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| 7. SceneCraft: Layout-Guided 3D Scene Generation ↗ | NeurIPS 2024 |
| 8. Contrastive Learning Relies More on Spatial Inductive Bias Than Supervised Learning: An Empirical Study ↗ | ICCV 2023 |
| 9. NeuralEditor: Editing Neural Radiance Fields via Manipulating Point Clouds ↗ | CVPR 2023 |
| 10. PointTree: Transformation-Robust Point Cloud Encoder with Relaxed K-D Trees ↗ | ECCV 2022 |
| 11. Is Self-Supervised Contrastive Learning More Robust Than Supervised Learning? ↗ | ICML 2022 Workshop |
| 12. TorchDrug: A Powerful and Flexible Machine Learning Platform for Drug Discovery ↗ | arXiv preprint, 2022 |
| 13. RNNLogic: Learning Logic Rules for Reasoning on Knowledge Graphs ↗ | ICLR 2021 |
| 14. Pre-training Assisted Scene-aware Dialogue Generation ↗ | DSTC8 Workshop at AAAI 2020 |

Earlier Experience

Mila – Quebec AI Institute

Research Intern; knowledge-graph reasoning with Prof. Jian Tang

Feb. – Aug. 2020

Montreal, Canada

Baidu, Natural Language Processing Department

Research / Development Intern; scene-aware dialogue generation (DSTC8 top-3 system)

July 2019 – Jan. 2020

Beijing, China

Competitions, Leadership & Coursework

Programming Competitions: ICPC EC-Final 2018, Gold Medal (4th); ICPC Qingdao Regional 2018, Gold Medal (2nd); CCPC Harbin Regional 2017, Gold Medal; NOI 2016, Gold Medal.

Additional Distinctions: CTSC 2017 (7th); CTSC 2016 Gold (8th); NOI Winter Camp 2016 Gold (3rd).

Leadership & Problem Setting: Problem setter for NOI, CTSC, ICPC, and CCPC events; Vice Chair, Tsinghua Student Association of Algorithms and Competitions (2018–2019).

Selected Coursework (Generative Models): Deep Generative Models (A+); Generative Models for Biomedicine and Life Sciences (A).

Vision & 3D: 3D Vision (A+); Deep Learning for Computer Vision (A); Computer Vision (A); Computational Photography (A+).

Learning: Meta-Learning (A+); Machine Learning (A).